BUGMASTERS • TEAM 011-5

Buff-Bugger

A location-based, gamified platform for discovering, documenting, and sharing bug finds in Boulder.

Molly Dembo / Trent Clatanoff / Benjamin Davison / Amrit Hari

CSCI 3308 • Spring 2026 • CU Boulder

ELEVATOR PITCH

Bug-finding as exploration, not data entry.

Buff-Bugger turns documenting local insects and arthropods into a game. Users report and categorize bugs they see while competing with friends on the leaderboard.

Built for:

- Bug enthusiasts and amateur entomologists
- Hikers, students, and outdoor hobbyists
- Anyone who wants to log finds and compete with friends

> *We wanted bug-finding to feel less like data entry and more like exploration, collection, and friendly competition.*
>
> — the BUGMASTERS team

THE PROBLEM

Logging biodiversity isn't fun.

Existing tools (iNaturalist, Seek)

- Strong scientific data logging
- Built around species identification
- Feels like filling out a form
- Minimal social / competitive layer
- Global focus, not locally social

Buff-Bugger

- Map-based local discovery
- Gamified collection + leaderboard
- Feels like exploration, not logging
- Built around sharing and competing
- Hyper-local: finds around Boulder

WHAT WE BUILT

Four core features

01 Interactive Map

Boulder-centered Google Map with clickable markers for every bug find. Click a marker to see common name, genus, and color.

02 Account & Session

Register, log in, and log out. Passwords hashed with bcryptjs; sessions persisted through express-session.

03 Bug Submission

Upload a photo, add location and a short description, and post. The find appears on the map and in the recent feed.

04 Social Layer

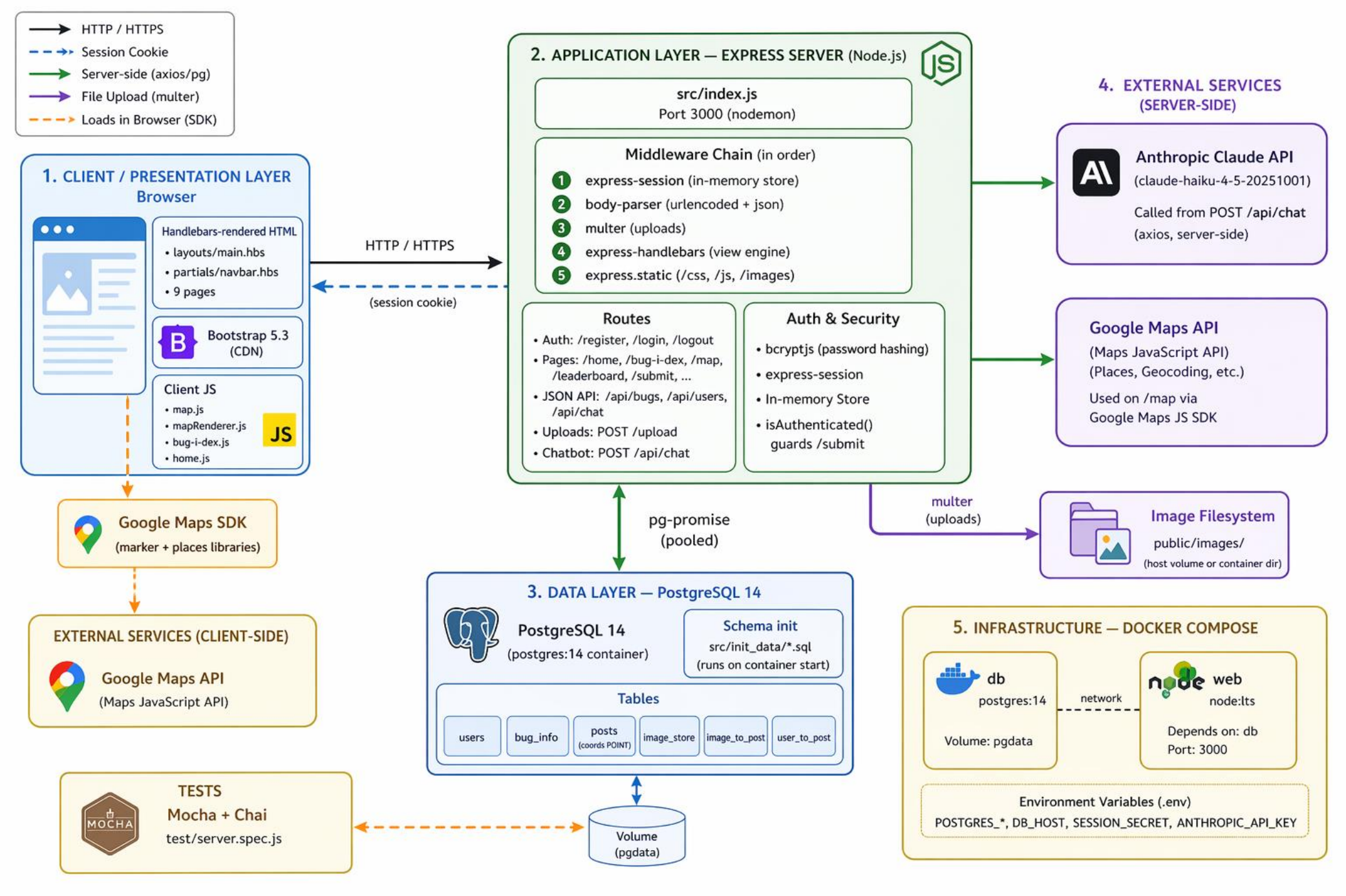
Leaderboard view to drive the friendly-competition framing of the app.

OUR STACK

Tools used + team rating out of 5.

Category	Tool	Purpose	Rating
Project Tracker	**GitHub Projects**	Kanban board for user stories and sprint tracking	●●●●○
Version Control	**GitHub**	Branching per feature; PRs reviewed before merge	●●●●●
Database	**PostgreSQL + pg-promise**	Users, posts, bug data, image-to-post relations	●●●●○
Application Server	**Node.js + Express**	Routing, sessions, validation, API	●●●●●
UI / Templating	**Handlebars**	Server-rendered pages + partials	●●●○○
Frontend Map	**Google Maps JS API**	Map rendering and bug marker display	●●●●●
File Upload	**multer**	Handling bug photo uploads	●●●●○
Auth	**bcryptjs + express-session**	Password hashing + persistent login	●●●●○
Testing	**Mocha + Chai + chai-http**	Server endpoint tests; registration positive + negative	●●●●○
Containerization	**Docker Compose**	Reproducible web + db environment	●●●●●
Deployment	**Render**	Hosted app + managed Postgres	●●●○○
Communication	**Discord + iMessage**	Async coordination + quick syncs	●●●●○

Rating reflects how useful the tool was for OUR project — not an absolute ranking.


HTTP / HTTPS
Session Cookie
Server-side (axios/pg)
File Upload (multer)
Loads in Browser (SDK)
1. CLIENT / PRESENTATION LAYER
Browser
Handlebars-rendered HTML
• layouts/main.hbs
• partials/navbar.hbs
• 9 pages
Bootstrap 5.3
(CDN)
Client JS
• map.js
• mapRenderer.js
• bug-i-dex.js
• home.js
JS
HTTP / HTTPS
(session cookie)
2. APPLICATION LAYER — EXPRESS SERVER (Node.js)
JS
src/index.js
Port 3000 (nodemon)
Middleware Chain (in order)
1 express-session (in-memory store)
2 body-parser (urlencoded + json)
3 multer (uploads)
4 express-handlebars (view engine)
5 express.static (/css, /js, /images)
Routes
• Auth: /register, /login, /logout
• Pages: /home, /bug-i-dex, /map, /leaderboard, /submit, ...
• JSON API: /api/bugs, /api/users, /api/chat
• Uploads: POST /upload
• Chatbot: POST /api/chat
Auth & Security
• bcryptjs (password hashing)
• express-session
• In-memory Store
• isAuthenticated() guards /submit
4. EXTERNAL SERVICES
(SERVER-SIDE)
Anthropic Claude API
(claude-haiku-4-5-20251001)
Called from POST /api/chat
(axios, server-side)
Google Maps API
(Maps JavaScript API)
(Places, Geocoding, etc.)
Used on /map via
Google Maps JS SDK
multer
(uploads)
Image Filesystem
public/images/
(host volume or container dir)
pg-promise
(pooled)
Google Maps SDK
(marker + places libraries)
EXTERNAL SERVICES (CLIENT-SIDE)
Google Maps API
(Maps JavaScript API)
3. DATA LAYER — PostgreSQL 14
PostgreSQL 14
(postgres:14 container)
Schema init
src/init_data/*.sql
(runs on container start)
Tables
users
bug_info
posts
(coords POINT)
image_store
image_to_post
user_to_post
Volume
(pgdata)
5. INFRASTRUCTURE — DOCKER COMPOSE
db
postgres:14
Volume: pgdata
network
web
node:lts
Depends on: db
Port: 3000
Environment Variables (.env)
POSTGRES_*, DB_HOST, SESSION_SECRET, ANTHROPIC_API_KEY
TESTS
Mocha + Chai
test/server.spec.js
MOCHA

Agile + Kanban, weekly cadence.

Approach

- Agile with a lean, scope-down mindset
- Kanban board on GitHub Projects
- User stories tied to issues
- Feature branches; PR review before merge

Cadence

- Weekly sprint, ~2-week feature arcs
- Weekly TA check-in (all 4 members)
- Rotating scribe for meeting minutes
- Release notes per sprint in /MilestoneSubmissions

Coordination

- Discord for async + code discussion
- iMessage for quick syncs
- Commits daily on feature branches
- Honest retros — scope vs. capacity

What worked: fast async coordination, clear PR ownership, weekly TA meetings kept scope honest.

What didn't: uneven week-to-week velocity; we over-scoped early and had to cut gamification depth.

Automated tests + real-user scenarios.

Automated (Mocha + Chai)

GET /welcome
Server reachability + JSON response shape

POST /register (positive)
Valid email + password → 200 + row created

POST /register (negative)
Invalid email → rejected with proper error

GET /api/bugs
Returns array of bug records for map

User Acceptance (UAT)

01 Register a new account
Username, password, confirm, valid email

02 Log in
Session persists; protected pages reachable

03 Create a bug post
Photo + location + short description → appears on map

04 View map + marker popup
Common name, genus, color visible

From UAT: *observing real users surface small friction points in the posting and map-viewing flows — several were addressed before final build.*

CHALLENGES

What broke, what we learned, what we cut.

01

Scope control

Problem: Original vision included full species unlocking, rarity-weighted scoring, and friend systems.

How we handled it: Cut the RPG-depth features to future scope. Kept the map, auth, submission, and leaderboard as the shippable core.

02

Integrating upload → DB → map

Problem: Photo upload, post creation, location capture, and marker rendering were developed on different branches and integrating them surfaced recurring merge and flow issues.

How we handled it: Coordinated integration sessions, agreed on a shared API shape for the map, and leaned on automated tests to catch regressions as features merged in.

03

Communication + load balance

Problem: Across four schedules, keeping task ownership clear and momentum steady week-to-week was harder than expected, and some features ended up bottlenecked on individual contributors.

How we handled it: Used the weekly TA meeting and the GitHub project board to keep ownership visible, and leaned on async channels to unblock each other between sync points.

FUTURE SCOPE

What we'd build with another semester.

01

Full species encyclopedia

Incorporate national species repository, extend location scope to all 50 states, current scope just Boulder

02

Rarity-weighted leaderboard

Score based on species rarity and habitat, not just raw find count. Weekly and seasonal leaderboards.

03

Stronger image classification

Automatic genus/species suggestions from the upload photo to lower friction and improve data quality.

04

Friends + social competition

Follow friends, compare collections, and issue challenges ("find 5 beetles this week").

05

Habitat-based achievements

Unlock badges for exploring specific ecosystems — foothills, wetlands, urban — and completing habitat sets.

06

Mobile-native experience

Camera-first mobile app; location capture on-the-go; push notifications for nearby rare finds.

Rubric requires at least 3 enhancements — we have 6.

LIVE DEMO

See Buff-Bugger in action.

https://www.loom.com/share/da4f705d00f04a92966fd552f6bf07cb

1	2	3	4	5	6
Land on home	**Click a marker**	**Register + log in**	**Submit a new bug**	**See it on the map**	**Leaderboard + AI**
Boulder map with existing markers	Popup: common name, genus, color	Session persists across pages	Photo + location + description	New marker renders immediately	Ranking view + GenAI feature

Thank you.

Buff-Bugger turns local bug-finding into a social, game-like experience — combining mapping, documentation, and competition in one place.

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Q & A